Haotian Zhang

hzhang44@usc.edu | Website | Google Scholar | GitHub

Education

University of Southern California Starting Spring 2027
Incoming undergraduate in Applied and Computational Mathematics.

American University of Paris August - December 2026
Undergraduate coursework; Fall 2026 semester.

Research Interests

Video reasoning; world models; applied mathematics.

I explore how models reason through visual representations, predict a changing world, and maintain a coherent understanding of objects and physical interactions.

Selected Honors and Awards

Hugging Face #1 Paper of the Week September 2026
For *Training Object Permanence in World Models*.

United States of America Mathematical Olympiad (USAMO) 2024
Qualifier; American Invitational Mathematics Examination (AIME): 13/15.

American Mathematics Competition (AMC 12) 2024
Global Top 1%.

S.-T. Yau High School Science Award - Mathematics 2024
Second Prize (Top 8), Southern Region Finals.

Berkeley Math Tournament 2023
National High Distinction (individual); national Top 2 overall and Top 3 in the Guts round.

High School Mathematical Contest in Modeling (HiMCM) 2024
Honorable Mention.

Publications

Preprint

Training Object Permanence in World Models

Haotian Zhang, Fengyuan Yu, Dezhi Luo, Haoran Sun, Zehong Zhao, Qingying Gao, ..., Yilun Du, Hokin Deng.

arXiv:2609.28654, 2026. Co-first author.

Paper | Project | Code

Research Experience

Hippocampal, Inc. June 2026 - Present
Research Intern

Training Object Permanence in World Models (WROP)

- Designed 150 Blender-based 3D task generators to study object permanence and solidity, supporting the project's 1.5-million-sample training corpus.
- Contributed to training a 16B-parameter video model, human evaluation, and manuscript preparation.
- Co-first authored the resulting preprint, recognized as Hugging Face's #1 Paper of the Week.

Research Experience (continued)

University of California, Santa Cruz June - August 2025
Research Intern, Summer Internship Program (SIP)

- Developed a generative-AI tool to scaffold research-paper reading through simulated reader-AI dialogue and iterative refinement of system prompts.
- Designed and evaluated prompts to help AI systems explain complex academic texts.
- Evaluated NotebookLM's Audio Overview for vocal tone, pacing, and content accuracy against professional podcast standards; explored prompt strategies for more natural narration.

Cambridge Centre for International Research (CCIR) June - September 2025
Research Student

- Completed coursework in Applied Linguistics & Big Data.
- Authored a research paper on the social functions of multilingualism along the ninth- and tenth-century Silk Road, combining linguistic theory with data-informed analysis.

Additional Academic Training

AwesomeMath Summer Program July 2024

- Completed Algebra 3.5 and Geometry 3, covering olympiad-level inequalities, functional equations, polynomials, and Euclidean geometry; wrote full-proof solutions with instructor feedback.

XAcademy Summer Program August 2024

- Completed natural language processing coursework and built an original game in a 24-hour capstone; received the Natural Symbiosis Theme Award.

International Youth Linguistics Winter Workshop February 2025

- Studied phonetics, phonology, morphology, syntax, semantics, and typology; solved Linguistics Olympiad problems through grammatical inference and proof-style reasoning.

Leadership and Teaching

Math Explorer Club, Shenzhen Middle School July 2024 - May 2026
President

- Coached students for AMC, AIME, and BMO contests through problem sets, mock exams, and full-solution discussions; taught calculus, linear algebra, combinatorics, number theory, and probability.

Mathematical Modeling Club, Shenzhen Middle School July 2024 - May 2026
President

- Taught mathematical modeling and its applications, along with Python and Java programming.

Words&Wonder Club, Shenzhen Middle School July 2024 - May 2026
Founder and President

- Designed linguistics and NLP learning activities, coached Olympiad problem-solving, and organized language walks exploring accessible and marginalized languages in public spaces.

Technical Skills

Programming: Python, Java. **Research tools:** Blender, synthetic 3D data generation, prompt engineering, human evaluation.